IT31043 - Mobile Application Development

Assignment: Mini Project – Final Report

Academic Year 2025 - 2026 | Third Year, First Semester

PCOS TRACKER

Polycystic Ovary Syndrome Health Management Application



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1. INTRODUCTION

1.1 About the Sustainable Development Goal (SDG)

Healthy lives and promoting wellbeing for all ages ensure the United Nations Sustainable Development Goal 3. It encompasses a broad range of health issues such as reproductive health, chronic disease prevention and universal access to healthcare services. [1] [1]

The most common hormone related disorders are Polycystic Ovary Syndrome (PCOS) experienced by females. Their reproductive years around the world with between 8 to 13% being affected and about 70% of patients not diagnosed due to lack of awareness. However, in Sri Lanka and throughout South Asia, cultural stigmatization related to reproductive issues, poor access to specialized gynecological treatment facilities and overall poor health literacy levels have exacerbated the issue.

SDG 3 targets are directly impacted by the condition:

- Target 3.1 - Reduce maternal mortality rate: PCOS increases the risk of conditions such as gestational diabetes and pre-eclampsia during pregnancy.
- Target 3.4 - Reduce premature mortality from NCDs: A major risk factor associated with type 2 diabetes, cardiovascular disorders and endometrial cancer considered by PCOS.
- Target 3.7 - Ensure universal access to healthcare services: Personal health monitoring, particularly where no specialists or financial means exist will be accessed by democratize A PCOS tracker application.
- Target 3.8 - Provide universal access to health services: Primary care provision to be extended to the population can allow Information technology solutions. [3] [4]

Women helping monitor their cycles, symptoms, diet, exercise and general mood through a mobile application by the PCOS Tracker application will assist in achieving these SDGs. [2] [1]

1.2 Problem (Background Study)

1.2.1 Problem Statement

Women suffering from Polycystic Ovary Syndrome (PCOS) are faced with great difficulties while trying to manage their multi-faceted and multi-system disease on a day to day basis. The existing general applications cannot cope with this problem due to an inability to cover all PCOS specific symptoms such as hormonal imbalances, irregular periods, body weight variations, high blood sugar levels, acne, hair loss etc. [4]

Thus, the lack of a special clinically driven application for PCOS leads to:

- The necessity to use multiple unrelated applications for period, diet or sport tracking purposes.
- Lack of an understanding of connections and relations between PCOS symptoms, diet and other aspects of lifestyle.
- Insufficient presentation of patients' health issues to doctors which negatively affects the consultation process.
- Psychological problems caused by a lack of appropriate information, support and community among those with PCOS. [3] [7]

1.2.2 Stakeholders & Their Needs

The following stakeholders can be considered for the PCOS Tracker app:

Stakeholders	Their Needs & Interests
Women suffering from PCOS (Primary Stakeholders)	Symptom recording, period tracking, food intake tracking, exercise tracking, mood tracking, medication reminders, and learning more about PCOS through various resources.
Gynecologists and Endocrinologists	Patient history records with symptoms recorded by the patient to make an accurate diagnosis and create custom treatments.
Nutritionists and Dietitians	Food intake records and nutritional information to plan meals with low glycemic index specifically for PCOS patients.
General Physicians	Reports about overall health conditions of women with PCOS for prompt referrals and future follow-ups.
Medical Institutes (Hospitals and Clinics)	Implications based on collected data to plan PCOS-specific programs.

Friends and Family Members Tools to learn more about PCOS and support PCOS sufferers. [6] [7]

1.2.3 Existing Solutions - Summary Analysis

There are several preexisting solutions which tackle the management of PCOS to some degree, yet each solution has certain weaknesses:

App Name	Functional Features	Weaknesses
Clue / Flo	Menstrual cycle tracking, ovulation tracking, symptoms recording	Does not correlate PCOS symptoms with menstrual cycles lacks diet and insulin monitoring.
MyFitnessPal	Nutrient, macro nutrient and calorie tracking, tracking exercises	No information on hormones or menstrual cycles non-medical solutions not focused on PCOS.
Ovia Health	Fertility management, some symptom monitoring	focused on fertility, does not provide PCOS oriented functionality such as androgen symptoms.
Bearable	Symptoms, mood and medication management	applicable for general use, extensive customization required no PCOS related education.

1.2.4 Scope and Limitations

The below mentioned scope and limitations guide the development process of the PCOS Tracker Application:

Scope: The PCOS Tracker application will be developed as a cross platform mobile application (for Android and iOS) alongside a web interface. Features that are considered core to the product include menstrual cycle tracker, symptoms tracker, food and nutrition tracker, exercise tracker, reminder notifications and reports on health status.

Time Limitation: The development process of this application is restricted within a period of one academic semester (about 16 weeks).

Budget Limitation: This application is developed as an academic activity and not funded commercially, hence all tools, frameworks, libraries and APIs used should only be free tier or open source.

Technical Limitation: Intermediate level proficiency in React Native and React.js is available in the development team. Back-end functionalities are to be implemented using Node.js with cloud-based databases (Firebase/Supabase).

Legal Limitation: Since this application deals with personal health information, the design, implementation and testing processes need to adhere to all laws related to data privacy protection such as GDPR Principles and Personal Data Protection Act (2022) - Sri Lanka. [6] [7]

1.3 Solution Overview

Concept Summary

PCOS Tracker is an exclusive mobile-based health management application made for those who have or are suspected of suffering from Polycystic Ovary Syndrome. It is a dedicated solution which incorporates all health related aspects of PCOS within a single intuitive application allowing the user to understand the syndrome better, track their progress and maintain contact with clinicians.

The main features of the solution include:

• **Track -** Logging of daily menstrual cycle, tracking of symptoms such as mood swings, fatigue, pain, meals consumed and exercise

• **Understand -** AI-driven analysis of correlations between various life aspects and symptoms, shown as graphical trends

• **Act -** Customized meal plans to suit PCOS conditions, custom exercise plans (low impact sports, yoga) meditation

• **Connect -** Ability to health reports for consultation with clinicians and optionally joining online communities to seek advice

The home page of the application includes (as seen in the screenshot attached):

- ➢ Current menstrual period information displayed in a summary form, current meal and exercise tracking
- ➢ The Quick Action menu buttons allow users to access the Calendar, Data Entry and Reports easily. The recent activity stream displays the most pertinent health activities entered by the user. [8]

2. REQUIREMENT ANALYSIS

2.1 Main Audience

The main target audience of the PCOS Tracker application consists of females and those of female biological sex aged between 15 to 45 years that have already been diagnosed with PCOS are under investigation for PCOS or think they might be suffering from some signs of PCOS. It includes teenagers suffering from irregular periods for the first time, young adults dealing with PCOS while also handling education or work issues and adult females suffering from PCOS while trying to conceive or nearing menopause.

Other secondary groups include healthcare professionals like gynecologists and GPs who will be able to access exported medical reports related to their patients and people from the support circle (family or partners) of those suffering from PCOS who might want to learn more about the disease from an educational perspective.

In terms of geographical distribution, the application targets people residing in South Asia (with emphasis on Sri Lanka) as its main market, while also catering to medium level computer literacy and lack of expertise in healthcare among other languages in the future. [7]

Persona 1: Undergraduate Student with Newly Diagnosed PCOS

Name	Nimali Perera, 21, Undergraduate Student - University of Colombo
Goals	Having trouble keeping track of her irregular periods and identifying causes for her acne, weight gain, etc. Desires to keep her PCOS under control naturally and learn more about PCOS-friendly dieting. Aims to prepare a symptom record to present to her gynecologist on her next visit.
Pain Points	Has difficulty remembering to keep up with logging her symptoms daily. Sees no relevance in current available applications since most are general and unrelated to her specific issue. Finds herself being flooded with too much information from all directions. Budget restrictions prevent frequent consultation with specialists.
Devices/OS	Samsung Galaxy A54 (Android 13), screen size - 6.4". Relies on mobile data and has access to home Wi-Fi only minimally.
Digital Literacy	Advanced - regularly uses social media and educational apps. Skilled at using mobile applications. Prefers simple interface rather than complicated statistics.
Accessibility Needs	Receives maximum benefit from high contrast mode due to astigmatism. Can potentially use an application in Sinhalese language. Requires reminder features with push notifications.

Persona 2: Working Professional Planning Pregnancy

Name	Dilrukshi Fernando, 31, Marketing Manager - Corporate Sector of Colombo
Goals	Keep track of ovulation cycles and fertility markers along with managing PCOS. Monitor how her exercises in the gym and low GI diet affect her menstrual cycle. Create reports that can be shared with her endocrinologist and fertility expert.
Pain Points	Very tight schedule, leaving little time for manual entry. Used several apps but not one combines tracking PCOS symptoms and fertility cycles. Needs exportable reports for doctor consultations. Privacy concerns.
Devices/OS	Apple iPhone 14 (iOS 17). 6.1" display. Very fast home and office Internet connection
Digital Literacy	High - highly experienced and tech-savvy professional working with productivity and health apps often.
Accessibility Needs	No assistive requirements. Prefers neat and professional interface. Strong privacy protection capabilities required, able to password protect her health records.

Persona 3: Teenage Student with Suspected PCOS

Name	Kavya Rajapaksa, 17, A/L Student - Kandy
Goals	Learn more about her menstrual issues to understand if they are normal or need any intervention. Learn about changes in her mood swings and the appearance of her acne. Get credible, yet easy-to-understand information on PCOS without being anxious about the process.
Pain Points	Ashamed to talk about her issues with family members, know nothing about PCOS. Unable to understand medical terms and feel intimidated by them. Little independence when it comes to her health and well-being. Concerns about privacy if the family accesses her phone.
Devices/OS	Huawei Nova 9 (Android 12). 6.57” screen. Mostly uses mobile data and sometimes connects to family Wi-Fi at home.
Digital Literacy	Intermediate - knows how to use social media and chatting apps but not health apps and productivity apps.
Accessibility Needs	Fairly easy to understand visual instructions with minimum texts. PIN required on app level for security purposes. Will be helpful if there’s bilingual Sinhala and English support.

2.2 Functional Requirements (FR)

Functional requirement specifies what needs to be done by the system or features required from the system.

	Functional Requirement
1	The splash screen displays PCOS Tracker branding and rotating health-related images before navigation.
2	Users can register, log in and access the PCOS Tracker mobile app.
3	Users can view a home dashboard with period, meal, workout, recent activity and profile summary.
4	Users can open profile settings, edit personal information, upload a profile image and save changes.
5	The home screen displays the saved user name as a greeting and shows the uploaded profile image.
6	Users can track period start date, end date, mood, pain level and special notes.
7	The period calendar highlights current, previous and predicted period ranges.
8	Users can view previous period history and predicted next period reminder.
9	Users can view a cycle insight report showing regular/irregular cycle information.
10	Users can access a period guide with cycle, bleeding, common symptoms, remedies, foods, hygiene, and doctor warning signs.
11	Users can access meal tracking from the home card, Add Today's Data, calendar legend, or tracking hub.
12	Users can track daily breakfast, lunch and dinner using checkboxes and save changes.
13	Users can view weekly and monthly meal tracking progress.
14	Users can open a Meal Gallery with PCOS-friendly weekly meal plans.
15	Users can view meal tracking reports with monthly meal completion data.
16	Users can open a Food Guide with healthy food choices and foods to avoid.
17	Users can access workout and wellness tracking for all, meditation, yoga and exercise.
18	Users can view daily wellness schedules with completion time and mark the selected activity as done.
19	Users can view work out details, duration, category and step sequence.
20	Users can view weekly and monthly workout wellness reports.
21	Users can open report hub screens for period, meal, workout and final report.
22	Users can view Privacy Policy from Settings.
23	Users can navigate between major screens using bottom navigation and back icons.

2.3 Non-Functional Requirements (NFR)

Non-functional requirements specify how well does the system fulfill its requirements or how well does it function.

Non-Functional Requirement	
Usability	The app uses clear cards, icons, bottom navigation, and back buttons for simple navigation.
Accessibility	Text uses readable sizes, strong contrast, and large tappable cards/buttons.
Performance	Local tracking data uses lightweight Shared Preferences for quick updates.
Reliability	Data entered in tracking screens remains available after leaving and reopening screens.
Security	User profile information is handled through Firebase Authentication /Firestore and local preference storage.
Maintainability	eatures are separated into Activities and store/helper classes such as PeriodStore, MealTrackingStore, and WorkoutTrackingStore
Consistency	Screens use a common light purple background, white cards, and consistent toolbar/back navigation
Scalability	The report hub and tracking hubs allow new report/tracking modules to be added later.

2.4 Technical Requirements

Technical requirements specify the technology used for building and running the system effectively.

Area	Requirement
Platform	Android mobile application
Language	Kotlin
UI	Android XML layouts, CardView, RecyclerView, BottomNavigationView, FloatingActionButton
Authentication	Firebase Authentication
Cloud Data	Firebase Firestore for user profile/check-in related data
Local Data	SharedPreferences through store classes for period, meal, workout, and profile data
Build Tool	Gradle / Android Gradle Plugin
Minimum Runtime	Android device/emulator capable of running Material Components and AndroidX
Assets	Drawable XML icons, JPG/PNG images, generated vector splash icons
IDE	Android Studio

3. DESIGN FOR MOBILE APPLICATION

3.1 Logo

The logo concept is that of a purple heart, which signifies care, women's health, feelings of well-being, and wellness. The application is aligned with SDG 3, which focuses on good health and well-being. It assists in tracking the following:

PCOS health data - Periods, Meals, Exercise, Yoga, Meditation

Applications for the Logo Concept:

- ❖ Application icon/logo
- ❖ Login screen logo
- ❖ Splash screen logo
- ❖ Profile and Wellness branding

Asset Files:

- app/src/main/res/drawable/ic_heart.xml
- app/src/main/res/drawable/ic_splash_yoga.xml
- app/src/main/res/drawable/ic_splash_exercise.xml
- app/src/main/res/drawable/ic_splash_meal.xml
- Imported images in app/src/main/res/drawable/

3.2 Theme Colors

The theme colors refer to the colors that are predominant in an application or system design in creating a consistent and appealing look.

Type	Color Role	Hex Code	Usage
Primary	Primary Purple	#8B7BB5	Buttons, selected tabs, active navigation, heart logo
Secondary	Teal	#4DB6AC	Meal tracking, yoga, completed states
Tertiary	Period Red	#E57373	Period cards, period dots, alerts
Other	Light Purple Background	#F0EEF8	Main screen background
	Purple Light	#EDE9F7	Icon backgrounds and badges

	Teal Light	#E0F5F1	Meal/yoga cards and icon backgrounds
	Red Light	#FFEBEE	Period icon background
	Green	#4CAF50 / #66BB6A	Normal cycle, success states
	Orange	#FFB74D	Medium/partial status
	Dark Text	#2D2D2D	Titles and primary text
	Body Text	#555555 / #666666	Paragraph and detail text
	Muted Text	#888888 / #AAAAAA	Captions, subtitles, inactive tabs
	Card White	#FFFFFF	Cards, inputs, bottom navigation

Sample Swatch:
Primary - #8B7BB5
Background - #F0EEF8
Teal - #4DB6AC
Period - #E57373
Text - #2D2D2D

3.3 Font Selection

The app employs the use of the default Android system sans-serif font. This is to ensure an appearance that looks native to Android as well as legibility.

UI Scenario	Font Type / Style	Color	Size
Main page titles	System sans-serif, Bold	#2D2D2D	24sp-28sp
Section/card titles	System sans-serif, Bold	#2D2D2D	16sp-20sp
Body text	System sans-serif, Regular	#444444 / #555555	13sp-15sp
Subtitles/captions	System sans-serif, Regular	#888888 / #AAAAAA	11sp-14sp
Button text	System sans-serif, Bold/All caps	#FFFFFF	14sp-16sp
Navigation labels	System sans-serif, Regular	Bottom nav tint colors	10sp-12sp
Report numbers	System sans-serif, Bold	Module colors	18sp-28sp

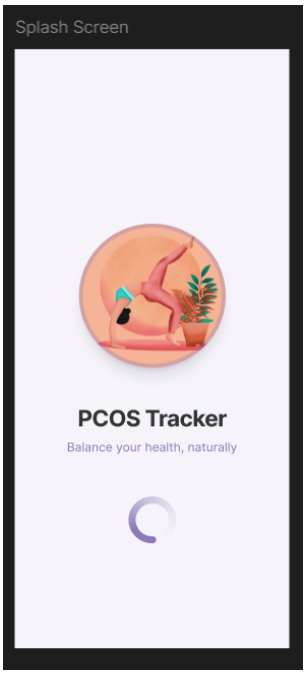
3.4 UI Prototype Designs

Design Tool: Android XML prototype implemented directly in Android Studio.

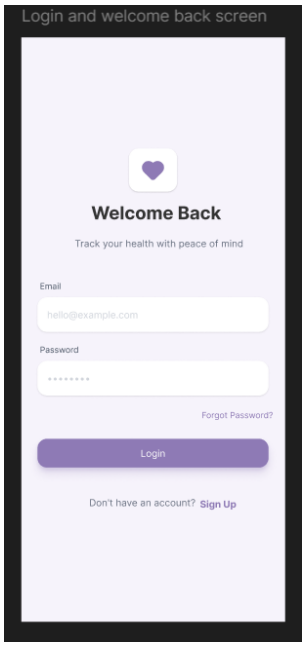
Proto Link:
https://www.figma.com/proto/VGDCrvrkN8t0AcnIZeKjVt/PCOS-Tracker?node-id=90-4275&starting-point-node-id=90%3A4275&show-proto-sidebar=1&t=V5aQYcnyCktFcAhC-1

Screenshots with Captions:

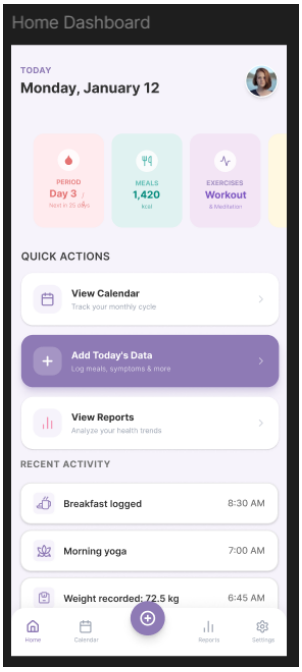
Splash Screen: Shows PCOS Tracker branding and rotating health images.



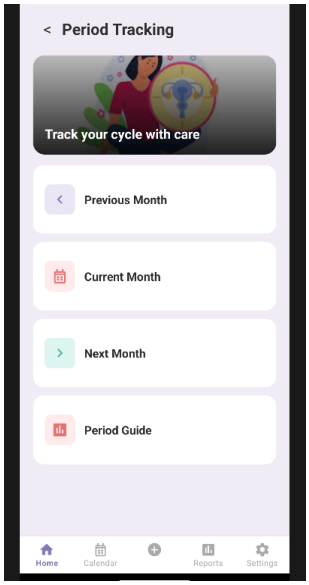
Login Screen: Allows user login with purple heart branding.



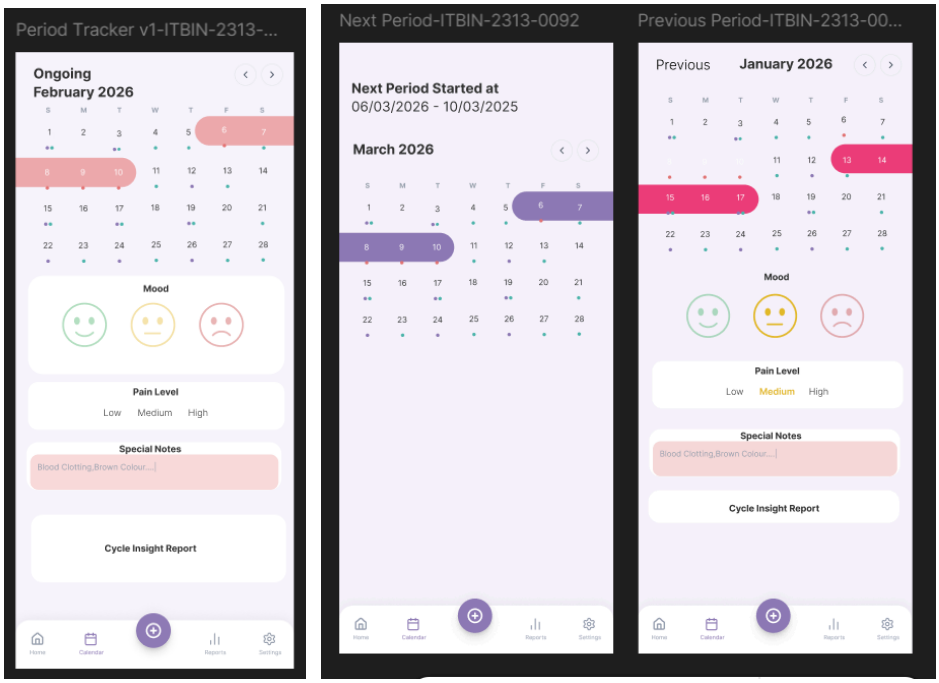
Home Dashboard: Shows period, meals, workout cards, quick actions and recent activity.



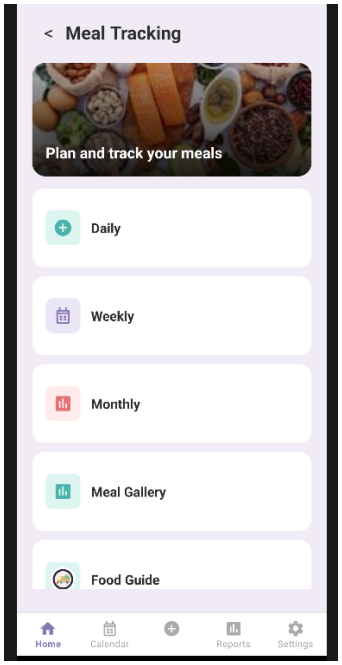
Period Tracking Hub: Navigation to previous/current/next month and period guide.



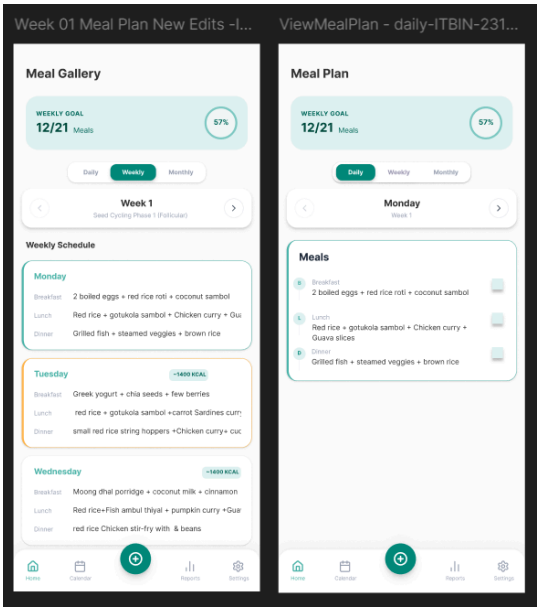
Period Calendar: Allows entering period dates, mood, pain level and notes.



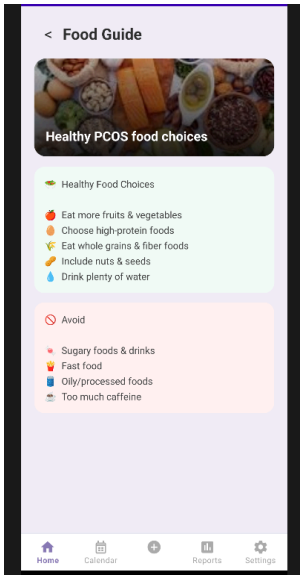
Meal Tracking Hub: Navigation to daily, weekly/monthly tracking, meal gallery and food guide.



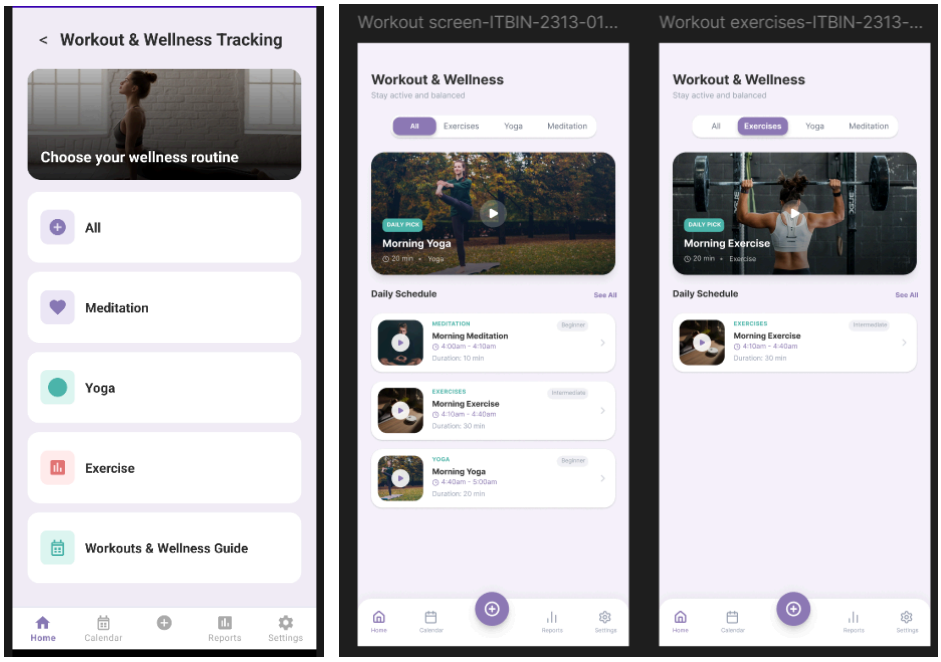
Daily Meal Tracking: Tracks breakfast, lunch and dinner completion.



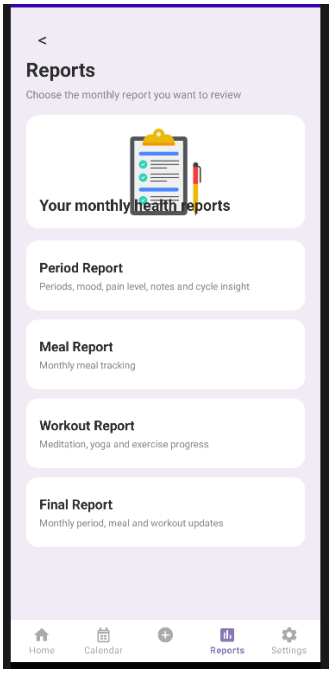
Food Guide: Shows healthy PCOS-friendly food choices and foods to avoid.



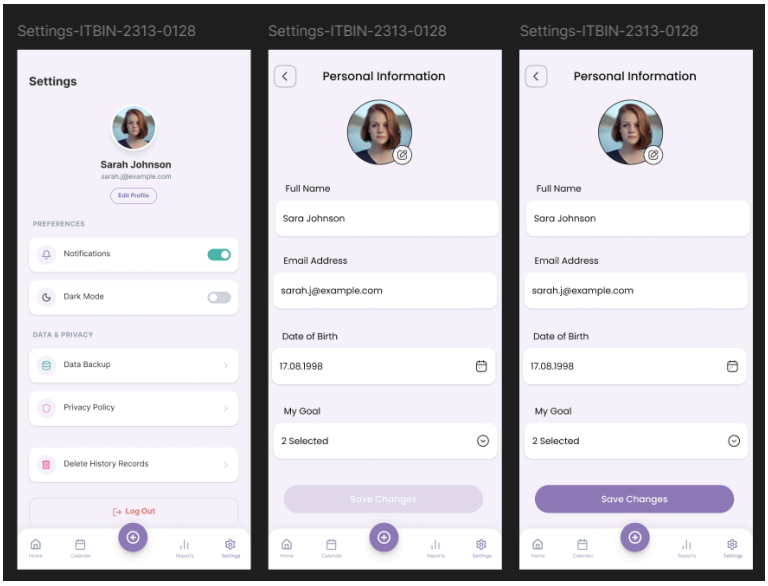
Workout & Wellness Hub: Navigation to all, meditation, yoga, exercise and guide



Reports Hub: Navigation to period, meal, workout and final reports.



Settings: Profile, preferences, privacy and account actions.



4. MOBILE APP DEVELOPMENT

4.1 Implemented Functions

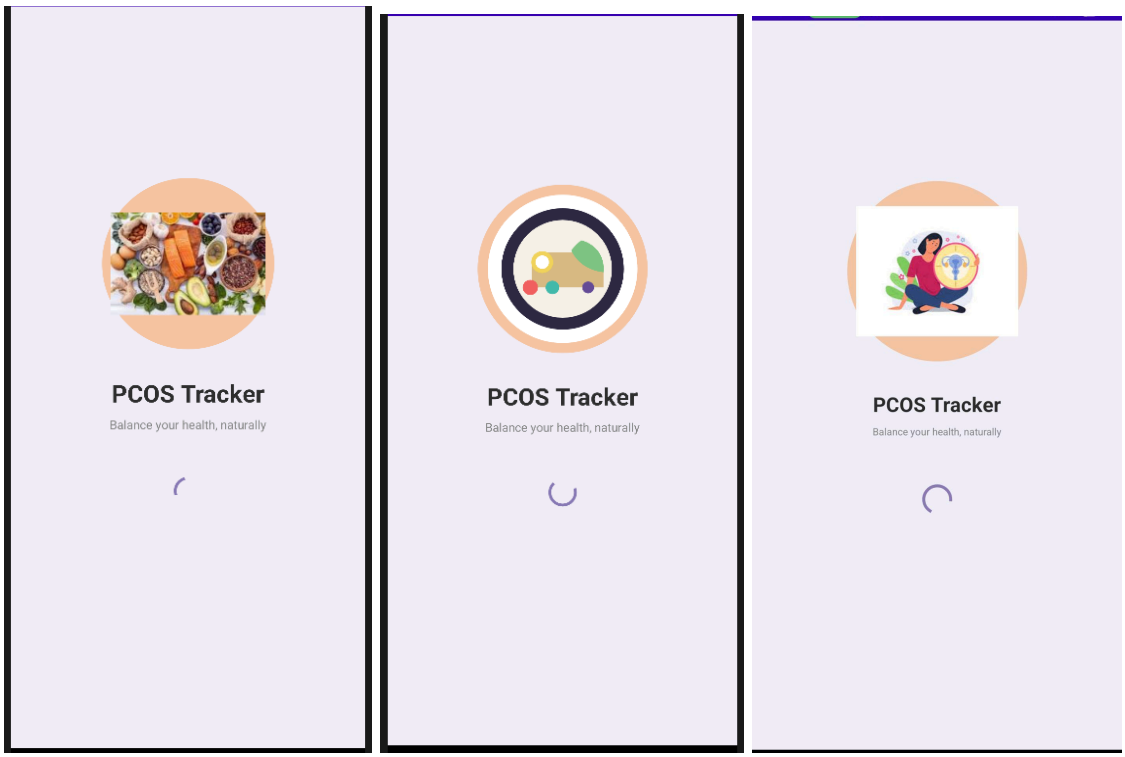
Module	Implemented Functions
Authentication	Login, sign-up, logout, authenticated navigation
Splash	Rotating start images, login/home routing
Profile	Edit name/email/DOB/goals, image upload, save profile, show profile on home/settings
Home	Summary cards, quick actions, recent activities, bottom navigation
Period	Track start/end date, mood, pain, notes, history, prediction, cycle insight
Calendar	Monthly overview with period, meal, workout dots and legend navigation
Meal	Daily meal tracking, weekly/monthly grid, meal gallery, food guide, meal report

Workout	Exercise/yoga/meditation routines, completion time, done status, detail pages
Reports	Period report, meal report, workout report, final report
Settings	Notifications, dark mode placeholder, data backup placeholder, privacy policy, delete history, logout
Privacy	Privacy Policy interface with protection principles

4.2 Mobile Interfaces (Screens)

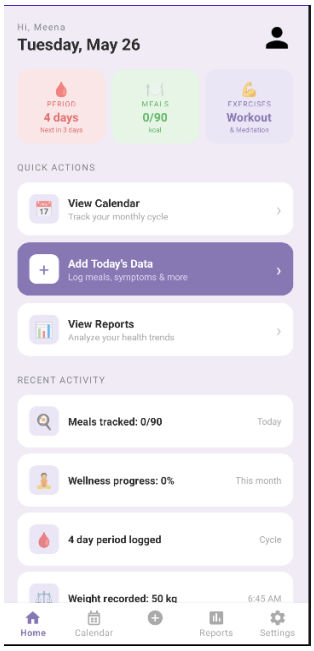
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1. Splash Screen
Purpose : Displays branding and rotating opening images before routing users.



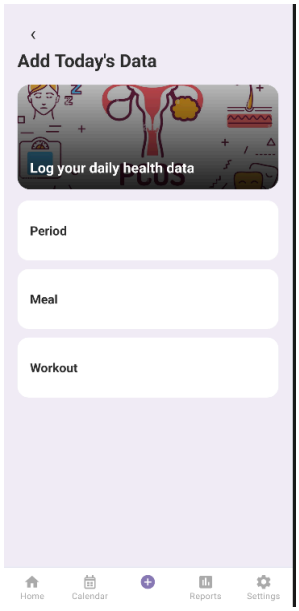
2. Home Dashboard

Purpose : Main overview of cycle, meals, exercise, quick actions, and recent activity.



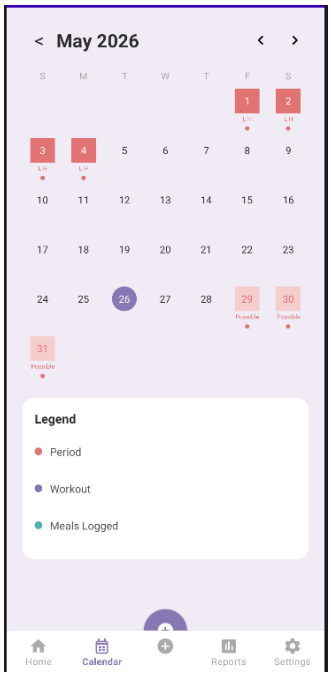
3. Add Today's Data

Purpose : Lets users choose Period, Meal, or Workout tracking.



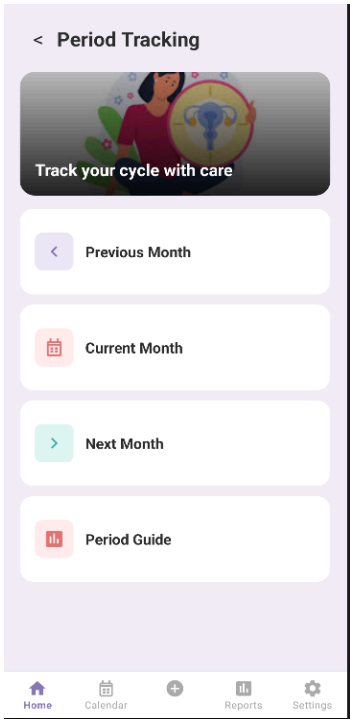
4. Calendar

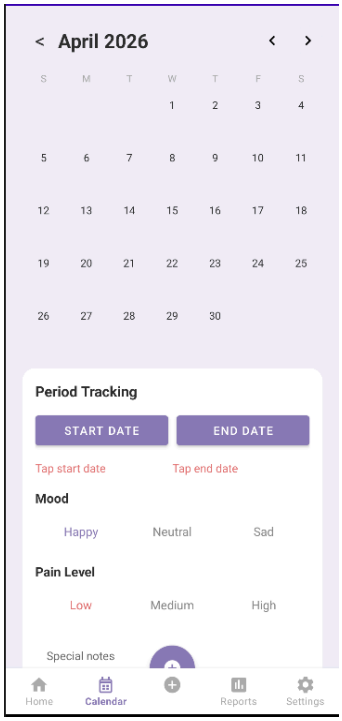
Purpose : Shows monthly tracking overview and period tracking calendar mode.

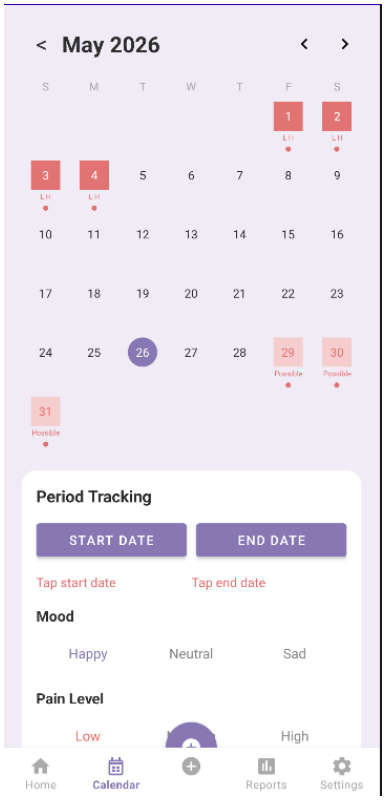


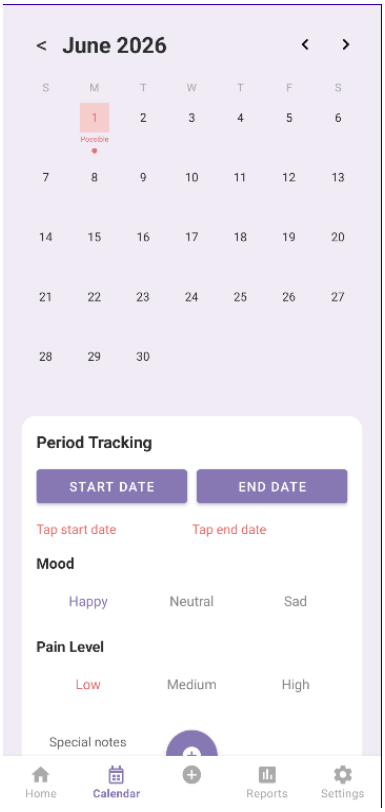
5. Period Tracking Hub

Purpose : Opens previous/current/next period month and period guide.



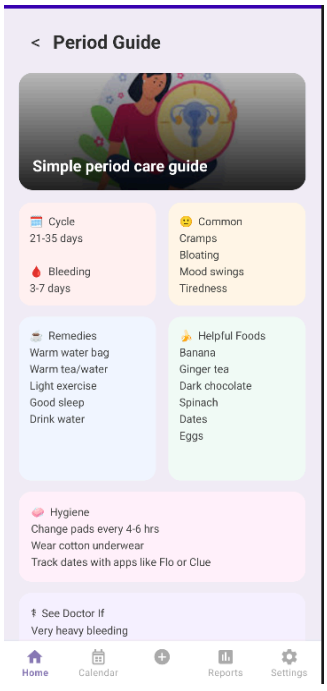






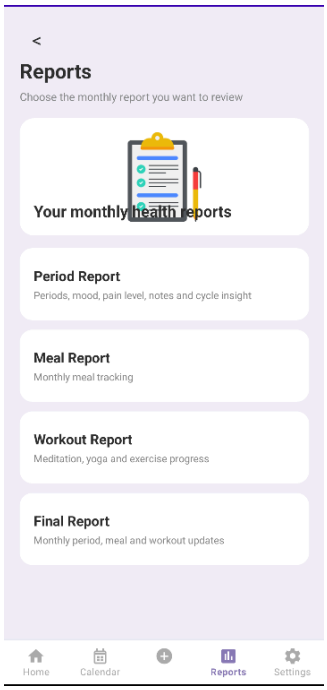
6. Period Guide

Purpose : Opens daily, weekly/monthly meal tracking, meal gallery and food guide.



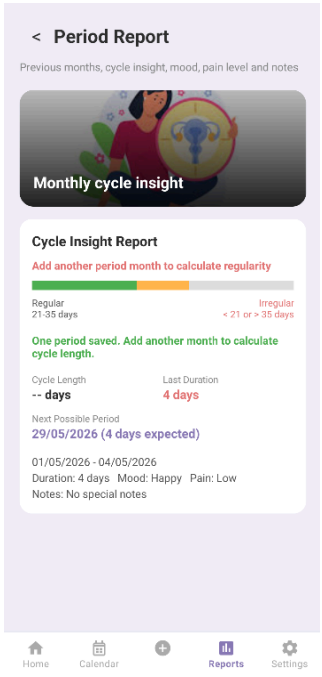
7. Reports Hub

Purpose: Opens period, meal, workout and final reports.



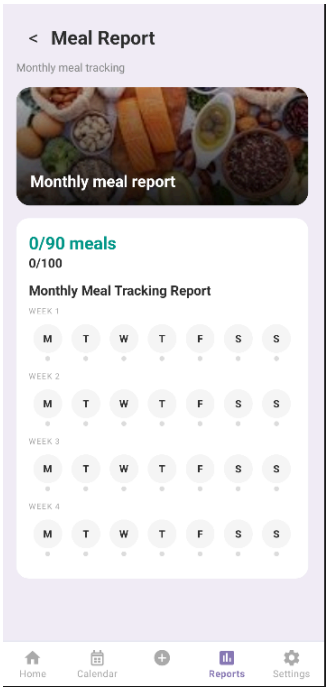
8. Period Report

Purpose: Shows period history and cycle insight report.



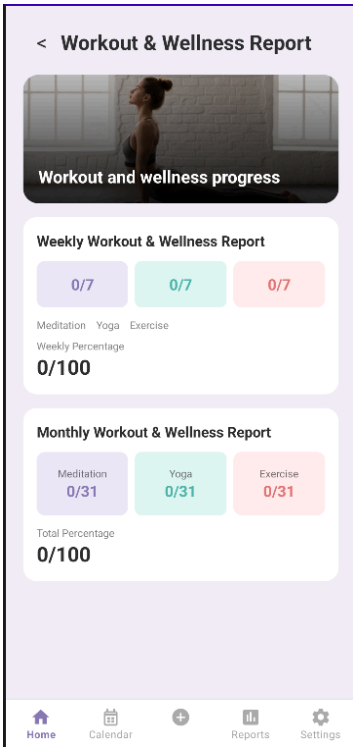
9. Meal Report

Purpose: Shows monthly meal tracking report.



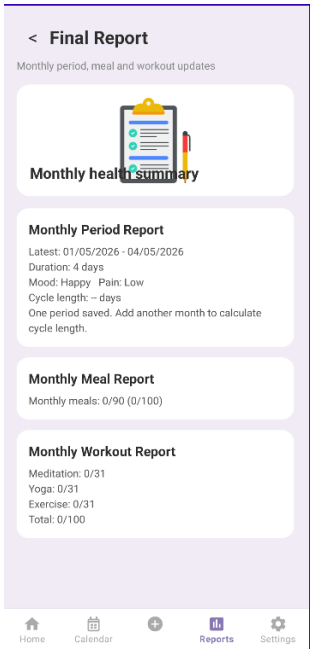
10. Workout Report

Purpose: Shows weekly/monthly meditation, yoga, and exercise report.



11. Final Report

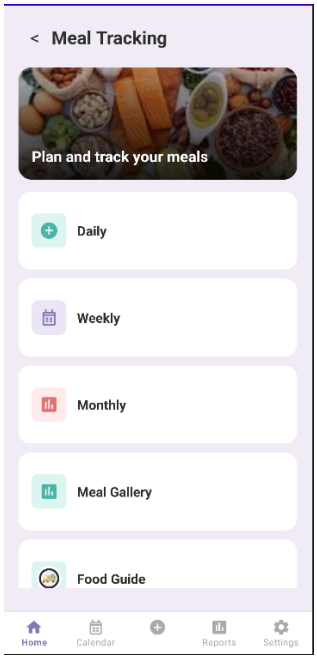
Purpose: Combines period, meal and workout monthly summaries.



ITBIN-2313-0088 - R G G M N Rajapaksha

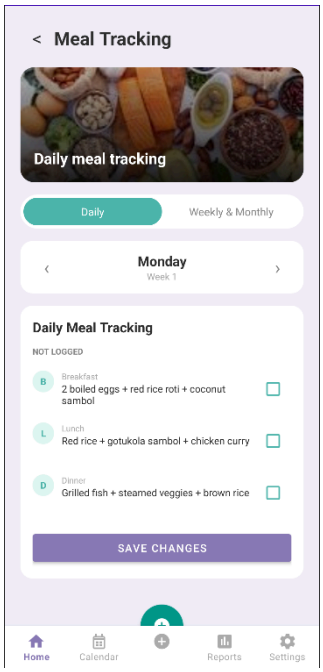
12. Meal Tracking Hub

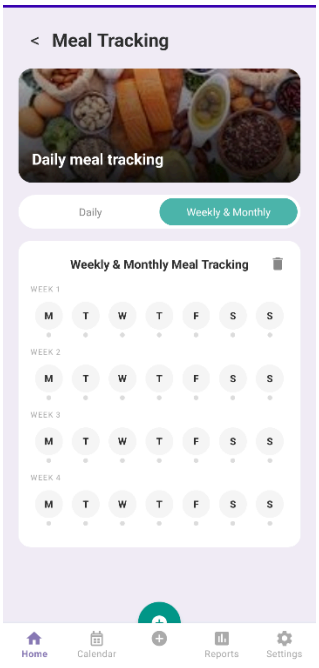
Purpose : Opens daily, weekly/monthly meal tracking, meal gallery and food guide.



13. Daily / Monthly Meal Tracking

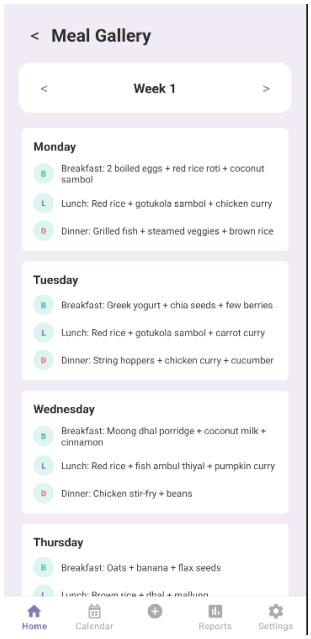
Purpose: Tracks meals and shows daily/weekly/monthly progress.





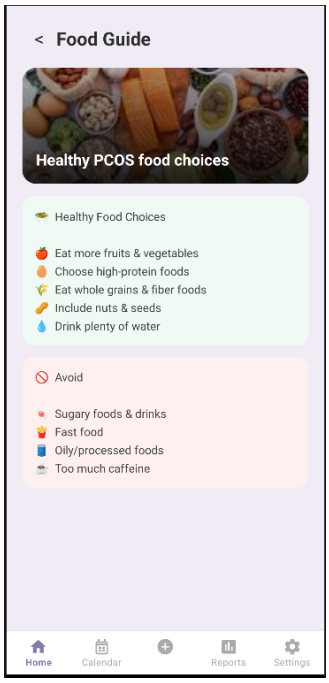
14. Meal Gallery

Purpose: Displays suggested PCOS friendly meal plans.



15. Food Guide

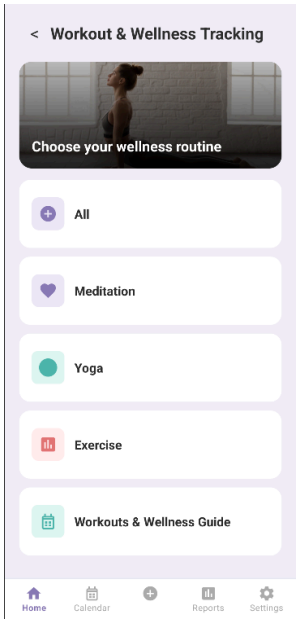
Purpose: Displays healthy food choices and foods to avoid for PCOS-friendly eating.



ITBIN-2313-0105 - D. Bhagya Senarathna

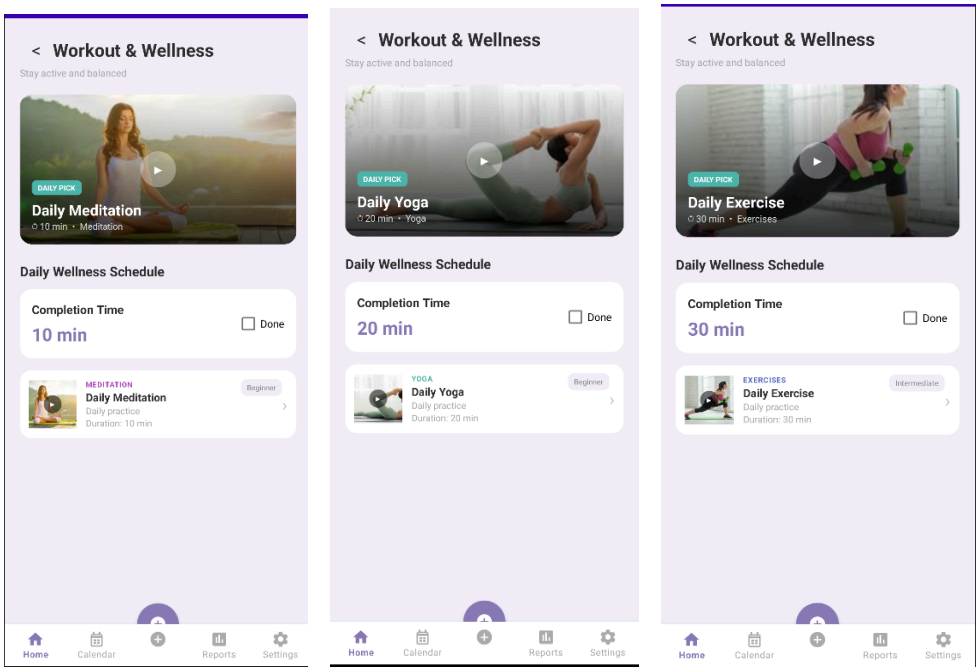
16. Workout & Wellness Hub

Purpose: Opens all, meditation, yoga, exercise and guide screens.



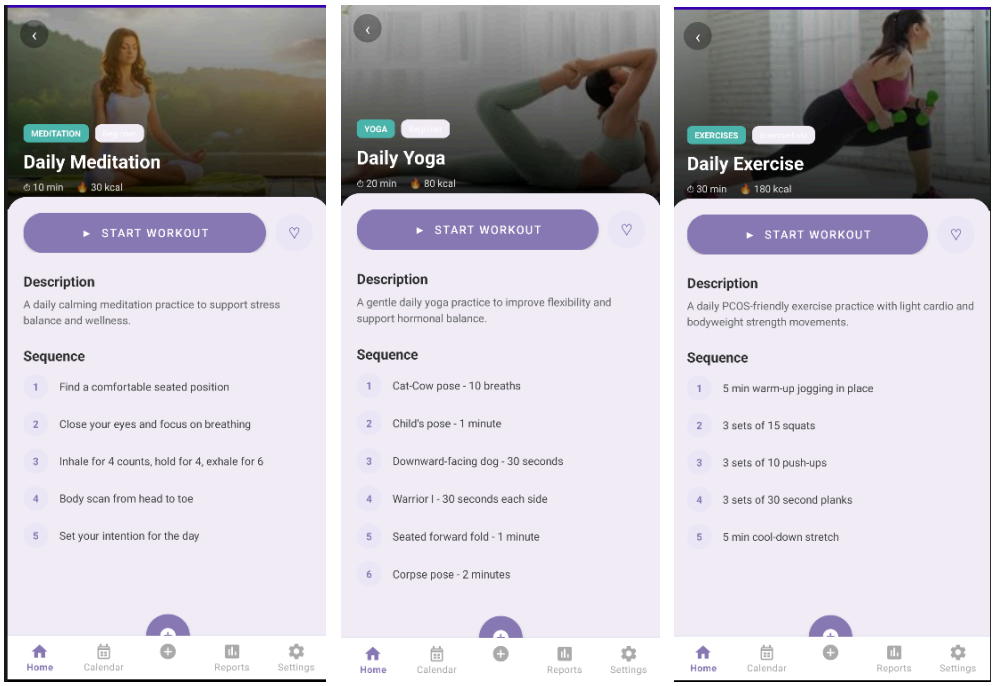
17. Workout & Wellness

Purpose: Shows selected daily wellness activity and schedule.



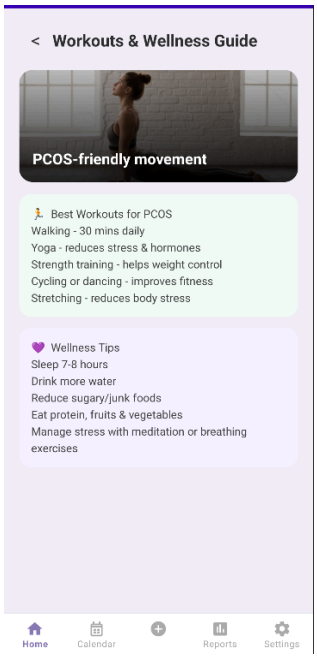
18. Workout Detail

Purpose: Shows routine details and step sequence.



19. Workout Guide

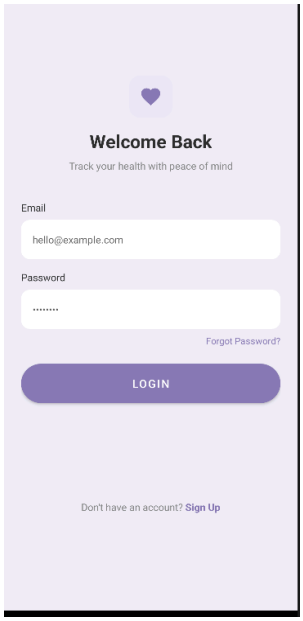
Purpose: Shows PCOS friendly workout and wellness tips.



ITBIN-2313-0128 - E N Sandeepani Wijerathne

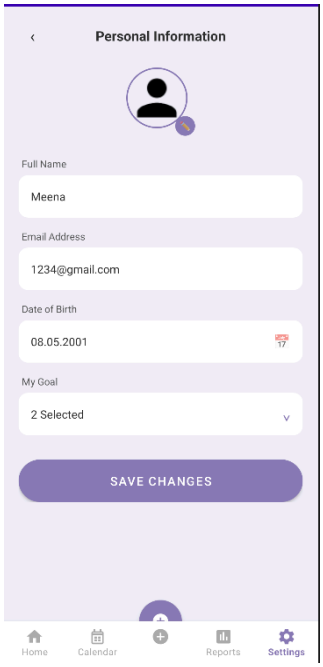
20. Login Screen

Purpose : Allows existing users to log in.



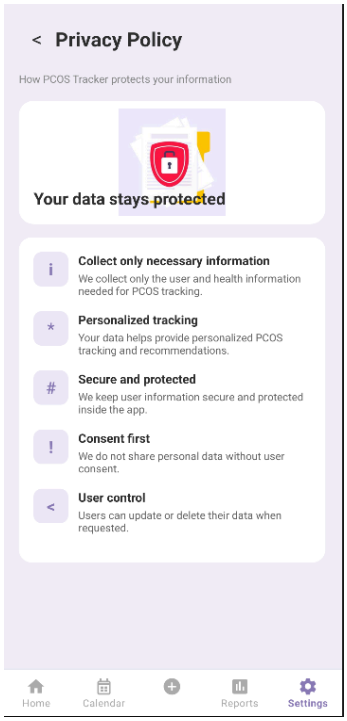
21. Personal Information

Purpose: Edits profile details and uploads profile image.

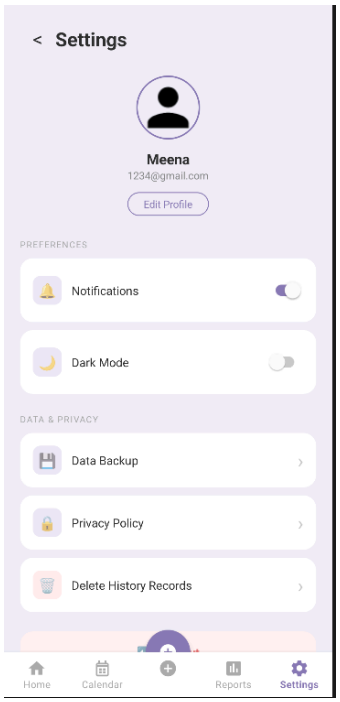


22. Privacy Policy

Purpose: Explains data collection, consent, protection and control.



23. Settings
Purpose: Manages profile, preferences, privacy, delete history, logout.



4.3 Code Segments (with Explanation)

This section includes important code segments used in every Activity of the PCOS Tracker mobile application. Each code segment is followed by a simple explanation describing its purpose and functionality.

Full activity source files are located in:

app/src/main/java/com/example/pcostracker/

ITBIN-2313-0092 - R G S M Rathwella

SplashActivity

Displays rotating splash images and routes to login or home based on Firebase user
Source File - SplashActivity.kt

```
override fun onCreate(savedInstanceState: Bundle?) {
```

```
super.onCreate(savedInstanceState)
setContentView(R.layout.activity_splash)

auth = FirebaseAuth.getInstance()
val splashImage = findViewById<ImageView>(R.id.splashImage)
val splashImages = listOf(
    R.drawable.opening_image_one,
    R.drawable.opening_image_two,
    R.drawable.ic_splash_meal
)
val handler = Handler(Looper.getMainLooper())
splashImages.forEachIndexed { index, image ->
    handler.postDelayed({
        splashImage.setImageResource(image)
    }, index * 800L)
}
```

MainActivity

Loads dashboard summaries, profile greeting/image, cards, recent activity and navigation. The following code displays the date on the dashboard of the home screen.

Source File - MainActivity.kt

```
private fun setupDate() {
val dateFormat = SimpleDateFormat("EEEE, MMMM d", Locale.getDefault())
val currentDate = dateFormat.format(Date())
findViewById<TextView>(R.id.currentDate).text = currentDate
}
```

DailyCheckInActivity

Legacy check-in screen for meals, period pain and workout duration. This method helps in saving the daily user input in Firebase Firestore. Pain intensity, menstrual cycle, and exercise type selected on a particular date are saved.
Source File - DailyCheckInActivity.kt

```
private fun saveToFirestore() {
val userId = auth.currentUser?.uid ?: return
val checkInData = hashMapOf(
"date" to today,
"painLevel" to selectedPainLevel,
"cycleType" to selectedCycleType,
"workouts" to selectedWorkouts
)
db.collection("users")
.document(userId)
.collection("checkIns")
.document(today)
.set(checkInData)
}
```

AddDataHubActivity

Provides Period, Meal and Workout tracking entry points. This activity serves as the main component in which the user's information is entered. Upon tapping the card, the app will take the user to another module such as period tracking, food tracking or workouts.
Source File - AddDataHubActivity.kt

```
class AddDataHubActivity : AppCompatActivity() {
override fun onCreate(savedInstanceState: Bundle?) {
super.onCreate(savedInstanceState)
setContentView(R.layout.activity_add_data_hub)
```

```
findViewById<CardView>(R.id.cardAddPeriod).setOnClickListener {
startActivity(Intent(this, PeriodTrackingHubActivity::class.java))
}
findViewById<CardView>(R.id.cardAddMeal).setOnClickListener {
startActivity(Intent(this, MealTrackingHubActivity::class.java))
}
findViewById<CardView>(R.id.cardAddWorkout).setOnClickListener {
startActivity(Intent(this, WorkoutWellnessHubActivity::class.java))
}
}
}
```

AddMealActivity

This code stores the entered meal details to Firebase Firestore. Meal type, description, selected day, and whether or not the task is completed are some of the fields saved in the database.
Source File - AddMealActivity.kt

```
private fun saveMeal(spinner: Spinner, dayNames: List<String>) {
val userId = auth.currentUser?.uid ?: return
val meal = hashMapOf(
"type" to mealType,
"description" to description,
"day" to selectedDay,
"completed" to false
)
db.collection("users").document(userId)
.collection("mealPlans")
.add(meal)
}
```

CalendarActivity

Builds monthly calendar, handles period tracking, date selection, history and legend navigation. This script allows saving the data about the menstrual cycle such as start date, end date, mood, pain intensity and remarks. Prior to saving the data, validation is done.
Source File - CalendarActivity.kt

```
private fun savePeriodLog() {
val start = selectedStartDate
val end = selectedEndDate
if (start == null || end == null) {
Toast.makeText(this, "Select period start and end dates", Toast.LENGTH_SHORT).show()
return
}

val notes = findViewById<EditText>(R.id.specialNotesInput)
.text.toString().trim()

PeriodStore.saveLog(
this,
PeriodLog(start, end, selectedMood, selectedPainLevel, notes)
)
}
```

PeriodTrackingHubActivity

Open period month modes and guide. This function opens the calendar screen with month details passed for period tracking.
Source File - PeriodTrackingHubActivity.kt

Period save function:

```
private fun savePeriodLog() {
    val start = selectedStartDate ?: return
```

```
    val end = selectedEndDate ?: start
    PeriodStore.saveLog(this, PeriodLog(start, end, selectedMood, selectedPainLevel, notes))
    setupCalendar()
    updateHistory()
}
```

This stores period tracking details and refreshes calendar/history.

```
private fun openPeriodMonth(monthOffset: Int) {
startActivity(
Intent(this, CalendarActivity::class.java)
.putExtra("periodMode", true)
.putExtra("monthOffset", monthOffset)
)
}
```

PeriodGuideActivity

Displays period education guide. The following code enables the user to go back to the previous screen from the period guide screen.
Source File - PeriodGuideActivity.kt

```
findViewById<TextView>(R.id.backBtn)
.setOnClickListener { finish() }
```

ReportHubActivity

Routes to period, meal, workout and final reports. This code helps navigate the user from the reports hub to the meals report screen.
Source File - ReportHubActivity.kt

```
findViewById<CardView>(R.id.cardMealReport)
.setOnClickListener {
startActivity(Intent(this, MealReportActivity::class.java))
}
```

ReportsActivity

Displays cycle insight and period history.
Source File - ReportsActivity.kt

```
private fun bindPeriodReport() {
    val logs = PeriodStore.getLogs(this)
    val averageCycle = PeriodStore.averageCycleLength(logs)?.toString() ?: "--"
    val latestDuration = logs.firstOrNull()?.durationDays?.toString() ?: "--"
    val nextPeriod = PeriodStore.nextPeriodStart(this)?.let { PeriodStore.formatDisplay(it) } ?: "Add
period history"
    val predicted = PeriodStore.predictedNextLog(this)
    val averageCycleValue = PeriodStore.averageCycleLength(logs)
    val isRegular = averageCycleValue != null && averageCycleValue in 21..35
    val statusText = when {
       logs.size < 2 -> "Add another period month to calculate regularity"
       isRegular -> "Your cycle is regular"
       else -> "Your cycle is irregular"
    }

    findViewById<TextView>(R.id.reportCycleLength).text = "$averageCycle days"
    findViewById<TextView>(R.id.reportPeriodDuration).text = "$latestDuration days"
    findViewById<TextView>(R.id.reportNextPeriod).text = predicted?.let {
       "$nextPeriod (${it.durationDays} days expected)"
    } ?: nextPeriod
    findViewById<TextView>(R.id.cycleInsightText).text = PeriodStore.insight(logs)
    findViewById<TextView>(R.id.cycleStatusText).text = statusText
    findViewById<TextView>(R.id.cycleStatusText).setTextColor(
       Color.parseColor(if (isRegular) "#4CAF50" else "#E57373")
    )
```

MealReportActivity

Displays monthly meal completion report. The following code shows the count of finished meals in the meal report section.
Source File - MealReportActivity.kt

```
private fun bindReport() {
val completed = MealTrackingStore.completedCount(this)
findViewById<TextView>(R.id.mealReportGoal).text =
"$completed/90 meals"
}
```

WorkoutWellnessReportActivity

Calculates and displays weekly/monthly workout completion.
Source File - WorkoutWellnessReportActivity.kt

```
private fun bindReport() {
    val days = WorkoutTrackingStore.daysInCurrentMonth()
    findViewById<TextView>(R.id.meditationWeeklyCount).text =
      "${WorkoutTrackingStore.weekCount(this, "Meditation")}/7"
    findViewById<TextView>(R.id.yogaWeeklyCount).text =
      "${WorkoutTrackingStore.weekCount(this, "Yoga")}/7"
    findViewById<TextView>(R.id.exerciseWeeklyCount).text =
      "${WorkoutTrackingStore.weekCount(this, "Exercises")}/7"
    findViewById<TextView>(R.id.workoutWeeklyPercent).text =
      "${WorkoutTrackingStore.weeklyPercent(this)}/100"
    findViewById<TextView>(R.id.meditationMonthlyCount).text =
      "${WorkoutTrackingStore.monthCount(this, "Meditation")}/$days"
    findViewById<TextView>(R.id.yogaMonthlyCount).text =
      "${WorkoutTrackingStore.monthCount(this, "Yoga")}/$days"
    findViewById<TextView>(R.id.exerciseMonthlyCount).text =
      "${WorkoutTrackingStore.monthCount(this, "Exercises")}/$days"
```

```
        findViewById<TextView>(R.id.workoutMonthlyPercent).text =
            "${WorkoutTrackingStore.totalPercent(this)}/100"
    }
    private fun setupNavigation() {
        val bottomNav = findViewById<BottomNavigationView>(R.id.bottomNav)
        bottomNav.setOnItemSelectedListener { item ->
            when (item.itemId) {
                R.id.nav_home -> { startActivity(Intent(this, MainActivity::class.java)); true }
                R.id.nav_calendar -> { startActivity(Intent(this, CalendarActivity::class.java)); true }
                R.id.nav_add -> { startActivity(Intent(this, AddDataHubActivity::class.java)); true }
                R.id.nav_reports -> { startActivity(Intent(this, ReportHubActivity::class.java)); true }
                R.id.nav_settings -> { startActivity(Intent(this, SettingsActivity::class.java)); true }
                else -> false
            }
        }
    }
}
```

FinalReportActivity

Combines monthly period, meal and workout summaries. The following code determines how far the user has come in their meal tracking process and shows this information on the final report screen.
Source File - FinalReportActivity.kt

```
private fun bindReport() {
val completedMeals = MealTrackingStore.completedCount(this)
val mealPercent = (completedMeals * 100) / 90
findViewById<TextView>(R.id.finalMealSummary).text =
"Monthly meals: $completedMeals/90 ($mealPercent/100)"
}
```

MealTrackingHubActivity

Opens daily tracking, monthly tracking, meal gallery and food guide. The following code loads the meal plan screen when the user chooses to track their meals every day.
Source File - MealTrackingHubActivity.kt

Daily meal tracking save:

```
saveChangesBtn.setOnClickListener {
    selectedMeals.forEach { mealId ->
        MealTrackingStore.setCompleted(this, mealId, true)
    }
    refreshDailyStatus()
}
```

This saves breakfast, lunch and dinner completion.

```
findViewById<CardView>(R.id.cardDailyMealTracking)
.setOnClickListener {
startActivity(Intent(this, MealPlanActivity::class.java))
}
```

ITBIN-2313-0088 - R G G M N Rajapaksha

MealPlanActivity

Handles daily meal checkboxes, save changes and monthly meal grid. The following function calculates the percentage completion of meal tracking.
Source File - MealPlanActivity.kt

```
private fun refreshProgress() {
val completed = MealTrackingStore.completedCount(this)
val percent = (completed * 100) / 90
findViewById<TextView>(R.id.progressText).text = "$percent%"
}
```

MealGalleryActivity

Displays weekly meal plans. This code updates the meal gallery screen by displaying the selected week.
Source File - MealGalleryActivity.kt

```
private fun bindWeek() {
```

```
findViewById<TextView>(R.id.currentWeekName).text =
"Week ${currentWeekIndex + 1}"
}
```

FoodGuideActivity

Displays food guide cards for healthy choices and foods to avoid. This code enables the user to go back to the previous screen from the Food Guide screen.
Source File - FoodGuideActivity.kt

Food Guide navigation:

```
findViewById<CardView>(R.id.cardFoodGuide).setOnClickListener {
    startActivity(Intent(this, FoodGuideActivity::class.java))
}
```

This code connects the Meal Tracking hub to the Food Guide interface.

ITBIN-2313-0105 - D Bagaya Senarathne

WorkoutWellnessHubActivity

Opens workout/wellness categories and guides. This code helps open the workout tracker screen if the user selects the workout tracking choice.
Source File - WorkoutWellnessHubActivity.kt

Workout category filtering:

```
val filtered = when (key) {
    "exercise" -> allWorkouts.filter { it.category == "Exercises" }
    "yoga" -> allWorkouts.filter { it.category == "Yoga" }
    "meditation" -> allWorkouts.filter { it.category == "Meditation" }
    else -> allWorkouts
}
```

This changes the workout screen based on the selected interface.

```
findViewById<CardView>(R.id.cardWorkoutTracker)
.setOnClickListener {
startActivity(Intent(this, WorkoutTrackerActivity::class.java))
}
```

WorkoutWellnessActivity

Filters all/meditation/yoga/exercise and updates completion time/status.
Source File - WorkoutWellnessActivity.kt

```
override fun onCreate(savedInstanceState: Bundle?) {
    super.onCreate(savedInstanceState)
    setContentView(R.layout.activity_workout_wellness)

    setupScheduleList(allWorkouts)
    setupDailySchedule()
    setupTabs()
    setupNavigation()

    findViewById<TextView>(R.id.backBtn).setOnClickListener { finish() }
    findViewById<FloatingActionButton>(R.id.fabAdd).setOnClickListener {
        startActivity(Intent(this, AddDataHubActivity::class.java))
    }
}
```

WorkoutDetailActivity

Displays workout image, routine info, and sequence steps.
Source File - WorkoutDetailActivity.kt

```
override fun onCreate(savedInstanceState: Bundle?) {
    super.onCreate(savedInstanceState)
    setContentView(R.layout.activity_workout_detail)
```

```
    val workoutId = intent.getStringExtra("workoutId") ?: "2"
    val workout = allWorkouts.find { it.id == workoutId } ?: allWorkouts.first()

    bindWorkout(workout)
    setupNavigation()

    findViewById<TextView>(R.id.backBtn).setOnClickListener { finish() }

    findViewById<Button>(R.id.startWorkoutBtn).setOnClickListener {
        Toast.makeText(this, "Starting ${workout.title}!", Toast.LENGTH_SHORT).show()
    }
```

WorkoutWellnessGuideActivity

Displays PCOS workout and wellness guidance.
Source File - WorkoutWellnessGuideActivity.kt

```
override fun onCreate(savedInstanceState: Bundle?) {
    super.onCreate(savedInstanceState)
    setContentView(R.layout.activity_workout_wellness_guide)

    findViewById<TextView>(R.id.backBtn).setOnClickListener { finish() }
    setupNavigation()
  }
```

ITBIN-2313-0128 - E N S Wijerathane

LoginActivity

Handles login form validation and Firebase login. This code performs authentication of the user using Firebase Authentication. The user will be redirected to the dashboard screen if the authentication process is successful.
Source File - LoginActivity.kt

```
private fun loginUser(email: String, password: String) {
auth.signInWithEmailAndPassword(email, password)

.addOnCompleteListener(this) { task ->

if (task.isSuccessful) {

startActivity(Intent(this, MainActivity::class.java))

finish()

}

}

}
```

SignUpActivity

Handles registration and account creation. This code helps create a user account through Firebase Authentication and redirects the user to the main screen post successful registration.
Source File - SignUpActivity.kt

```
auth.createUserWithEmailAndPassword(email, password)

.addOnCompleteListener(this) { task ->

if (task.isSuccessful) {

startActivity(Intent(this, MainActivity::class.java))

finish()

}

}
```

PersonalInfoActivity

Handles profile input, image picker, local save and Firestore update. This code enables saving of the user's personal details like name, email address, birthday, and profile picture.
Source File - PersonalInfoActivity.kt

Profile image upload and save:

```
private val imagePicker = registerForActivityResult(ActivityResultContracts.OpenDocument()) { uri ->
    uri ?: return@registerForActivityResult
    selectedImageUri = uri
    contentResolver.takePersistableUriPermission(uri,
Intent.FLAG_GRANT_READ_URI_PERMISSION)
    findViewById<CircleImageView>(R.id.profileImageEdit).setImageURI(uri)
}
```

This code opens Android's document image picker, stores the selected URI and previews it in the profile image view.

Home greeting and profile image:

```
private fun setupProfileHeader() {
    val storedName = ProfileStore.name(this)
    findViewById<TextView>(R.id.todayLabel).text = "Hi, ${storedName.ifEmpty { "there" }}"
    val imageUri = ProfileStore.imageUri(this)
    if (imageUri.isNotEmpty()) {
        findViewById<CircleImageView>(R.id.profileImage).setImageURI(Uri.parse(imageUri))
    }
}
```

This displays the saved profile name and image on the home dashboard.

```
private fun setupSaveButton() {
findViewById<Button>(R.id.saveChangesBtn).setOnClickListener {

val name = findViewById<EditText>(R.id.fullNameInput)
```

.text.toString().trim()

ProfileStore.save(this, name, email, dob,

selectedImageUri?.toString())

}

}

PrivacyPolicyActivity

Binds privacy policy rows and handles settings navigation. This function helps load the privacy policy information on the screen.
Source File - PrivacyPolicyActivity.kt

Privacy Policy row binding:
private fun bindPolicyRow(rowId: Int, icon: String, title: String, body: String) {

  val row = findViewById<View>(rowId)

  row.findViewById<TextView>(R.id.privacyIcon).text = icon

  row.findViewById<TextView>(R.id.privacyTitle).text = title

  row.findViewById<TextView>(R.id.privacyBody).text = body

}

This keeps the privacy policy screen reusable and easy to update.

SettingsActivity

Loads profile, handles settings actions, privacy policy, delete history, logout. This function helps log out the currently logged-in user and takes him/her to the login screen.
Source File - settingsActivity.kt

findViewById<TextView>(R.id.logoutBtn)

.setOnClickListener {

FirebaseAuth.getInstance().signOut()

startActivity(Intent(this, LoginActivity::class.java))

finish()

}

5. MOBILE APP TESTING

5.1 Test Case Table

ITBIN-2313-0092 - R G S M Rathwella						
Test ID	**Feature**	**Preconditions**	**Steps**	**Expected Result**	**Actual**	**Status**
TC01	Splash navigation	App installed	Open app	Splash appears and navigates to login/home	Build verified	Pass
TC02	Home quick actions	App on Home	Tap View Calendar, Add Today's Data, View Reports	Correct hub/screen opens	Implemented	Pass
TC03	Period tracking hub	App open	Tap Period card	Period Tracking hub opens	Implemented	Pass
TC04	Period save	Period screen open	Select start/end, mood, pain, notes, save	Period appears in history/calendar	Implemented	Pass
TC05	Next period reminder	Period history exists	Open Period Report	Next period reminder appears	Implemented	Pass
TC06	Period guide	Period hub open	Tap Period Guide	Period guide opens with care information	Implemented	Pass
TC07	Period report	Reports hub open	Tap period Report	Cycle insight report appears	Implemented	Pass
TC08	Meal report	Reports hub open	Tap Meal Report	Monthly meal report appears	Implemented	Pass
TC09	Workout report	Reports hub open	Tap Workout Report	Weekly/monthl y report appears	Implemented	Pass

TC10	Final report	Reports hub open	Tap Final Report	Combined report appears	Implemented	Pass
TC11	Back navigation	Any sub-screen open	Tap back icon	Previous interface opens/closes current screen	Implemented	Pass
TC12	Bottom navigation	Any main screen open	Tap Home/Calendar/ Add/Reports/Settings	Correct screen opens	Implemented	Pass
TC13	Build test	Source code available	Run .\gradlew.bat :app:assembleDebug	Debug APK builds successfully	Build passed	Pass
ITBIN-2313-0088 - R G G M N Rajapaksha						
Test ID	**Feature**	**Preconditions**	**Steps**	**Expected Result**	**Actual**	**Status**
TC14	Meal hub	App open	Tap Meals card	Meal Tracking hub opens	Implemented	Pass
TC15	Daily meal tracking	Meal Tracking > Daily	Tick meals and tap Save Changes	Status updates to Done/Partial	Implemented	Pass
TC16	Weekly/monthly meal grid	Meal Tracking > Weekly & Monthly	Tap week day circle	Daily meal tracking opens for that day	Implemented	Pass
TC17	Meal gallery	Meal hub open	Tap Meal Gallery	Weekly meal plan list opens	Implemented	Pass
TC18	Food guide	Meal hub open	Tap Food Guide	Food Guide opens with healthy choices and avoid list	Implemented	Pass

ITBIN-2313-0105 - D Bagya Senarathne						
Test ID	**Feature**	**Preconditions**	**Steps**	**Expected Result**	**Actual**	**Status**
TC19	Workout hub	App open	Tap Workout card	Workout & Wellness Tracking hub opens	Implemented	Pass
TC20	Workout category	Workout hub open	Tap Meditation/Yoga/Exercise	Correct filtered interface opens	Implemented	Pass
TC21	Workout completion	Workout screen open	Tick Done	Completion is saved for selected activity	Implemented	Pass
ITBIN-2313-0128 - E N S Wijerathne						
Test ID	**Feature**	**Preconditions**	**Steps**	**Expected Result**	**Actual**	**Status**
TC22	Login	Existing user account	Enter email/password and tap Login	User opens Home screen	To test on emulator	Pending
TC23	Profile save	User logged in	pen Settings > Edit Profile, enter name, select image, save	Name/image saved and shown on Home	Implemented	Pass
TC24	Privacy policy	Settings open	Tap Privacy Policy	Privacy Policy interface opens	Implemented	Pass

5.2Appendix

Main Source Paths

- **Kotlin activities:** app/src/main/java/com/example/pcostracker/
- **XML layouts:** app/src/main/res/layout/
- **Drawable icons/images:** app/src/main/res/drawable/
- **Bottom navigation menu:** app/src/main/res/menu/bottom_nav_menu.xml
- **Manifest:** app/src/main/AndroidManifest.xml

6. CONCLUSION AND DISCUSSION

A mobile application called PCOS Tracker has been successfully created to enable women to control and track activities related to their PCOS health issues in an organized manner. The program incorporates a wide range of functions such as tracking periods, creating meals, planning workouts, tracking daily health status, reporting and user profile management in one application.

Use of the Firebase Authentication and Firebase Firestore resulted in providing secured user management services and cloud-based storage of the data collected by the application. Users are able to create accounts, login into their accounts, store their personal health information and access the data any time they need. Navigation between different activities is made smooth by use of Android Intents and Bottom Navigation.

The developed activities show how Android Studio and Kotlin languages could be utilized in developing useful and efficient healthcare applications. The calendar and report activities help in monitoring of menstrual cycles and health progress, while meal and workout activities make users adopt healthier lifestyles.

In the course of developing the application, various software engineering practices have been observed, which include modular programming, use of reusable code segments, user interface design, data validation and database incorporation.

First of all, the system's simplicity and clarity in the user interface allow for convenient application usage. Moreover, PCOS Tracker is a tool for individual health control where users can track their moods, pain level, nutrition data, physical exercises, and other aspects of personal wellbeing.

Nevertheless, the system has some limitations. First, all data are entered manually by users and, thus, there is no support for more advanced functions like AI suggestions, notifications, online consultations with doctors, and wearable devices.

To enhance the system and add some useful features, the application should be improved by implementing the following functionality:

Suggestions from AI about patients' health state

- Notifications and reminders
- Support for online consultations with doctors
- Integration with fitness trackers and smart watches
- Graphical analysis and reporting
- Language selection support
- Backup and synchronization with cloud services

Summing up, it should be noted that the PCOS Tracker application has fulfilled its main function and successfully tracks PCOS-related activities using digital technologies. This application serves as a good example of practical knowledge gained during the course of Android application development, Firebase and cloud storage application, as well as mobile healthcare systems.

GOOGLE DRIVE FOLDER LINK
Includes:

APK file -
https://drive.google.com/drive/folders/1hWesLDe2mF_nBRrC_xRRvhxwgpk16bbe?usp=sharing

Demo video link -
https://drive.google.com/drive/folders/1vFheqgqvsqCZLoP59NnT7jPuzTiXiUFM?usp=sharing

GitHub repository link - https://github.com/savindirathwella333-create/PCOSTracker/branches

Project report

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